

3D-Integrated SRAM Components for High-Performance Microprocessors

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Abstract—3D integration is an emergent technology that has the potential to greatly increase device density while simultaneously providing faster on-chip communication. 3D fabrication involves stacking two or more die connected with a very high density and low-latency interface. The die-to-die vias that comprise this interface can be treated as regular on-chip metal due to their small size (on the order of 1 μm) and high speed (sub-FO4 die-to-die communication delay). The increased device density and the ability to place and route in the third dimension provide new opportunities for microarchitecture design. In this paper, we focus on the 3D-integrated designs of SRAM structures. We show that the dense die-to-die vias enable 3D-integrated SRAM components that are partitioned at the level of individual wordlines or bitlines. This results in a wire length reduction within SRAM arrays, and a reduction in the area footprint, which reduces the wires required for global routing. The wire length reduction provides simultaneous latency and energy reduction benefits, e.g., 47 percent latency reduction and 18 percent energy reduction for a 4 MB 4-die stacked 3D SRAM array. A 3D implementation of a 128-entry multiported SRAM array achieves a 36 percent latency improvement with a simultaneous energy reduction of 55 percent. As planar designs adapt high-performance techniques such as hierarchical wordlines to improve performance, 3D integration provides even larger benefits, making it a desirable technology for high-performance designs. For the 4 MB SRAM array, the 3D-integrated designs provide additional latency reduction benefit over the planar designs when hierarchical wordlines are implemented in both planar and 3D designs.

Index Terms—3D-integrated technology, die stacking, SRAM arrays.

1 INTRODUCTION

THE semiconductor industry faces an increasing number of challenges that must be overcome to keep pace with Moore's Law and industry projections. Some of these challenges include poor scaling of wire RC delays, increasing power consumption, and manufacturing challenges. 3D integration has the potential to address these challenges for future high-performance microprocessors. 3D integration can greatly increase device density while simultaneously providing faster on-chip communication. 3D integration technologies are largely orthogonal to the planar processes that provide each of the individual die. Advances in planar and 3D integration complement each other. Therefore, 3D integration can reap the benefits of advances in traditional planar processes such as double-gate transistors [1], tri-gate transistors [2], finFETs [3], strained silicon [4], and metal gates [5]. Processor companies are actively researching the technology [6], [7], [8], [9] and the embedded domain is already producing products in 3D [10]. There has been a great research interest [6], [7], [11], [12], [13], [14], [15], [16], [17], [18], [19] in circuits and microarchitecture using the 3D integration technology. In addition to the research in circuits and microarchitecture, there has been a great deal of

activity in the computer-aided design (CAD) area [20], [21], [22]. In our work, we propose the 3D-integrated designs of various SRAM arrays with functional block partitioning that requires a redesign at the circuit level. To the best of our knowledge, our work is the first to unify the design ideas of large SRAM arrays such as caches and multiported SRAM arrays such as register files into a general 3D-integrated design framework for SRAM arrays. We also explore existing high-performance optimizations in planar circuit designs and demonstrate that the 3D-integrated circuits may provide more benefits even with an optimized planar baseline configuration.

The rest of the paper is organized as follows: Section 2 introduces the 3D technology. Section 3 describes the planar SRAM components. Section 4 describes the 3D-integrated designs of large, banked SRAM arrays. Section 5 describes the 3D-integrated designs of multiported SRAM arrays. Section 6 presents our results and analysis. Finally, Section 7 concludes the paper.

2 3D TECHNOLOGY

There are several proposed methods for vertically integrating multiple circuit layers such as Multilayer buried structures [23] and die bonding [24], [25]. Fig. 1a shows an MLBS 3D design. In the multilayer buried structures (MLBSs) 3D technology, multiple device layers are successively (sequentially) fabricated in a stacked fashion. Layer-to-layer connections are made from either interlayer vias or from direct source-drain/source-drain contacts. The advantage of MLBS technology is that the vertical 3D vias can potentially scale down with feature size due to the use of local polysilicon wires for connection. The die bonding

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Manuscript received 26 June 2007; revised 5 Nov. 2007; accepted 4 Dec. 2007; published online 30 June 2009.

Recommended for acceptance by A. Gonzalez.

For information on obtaining reprints of this article, please send e-mail to: tc@computer.org, and reference IEEECS Log Number TC-2007-06-0258. Digital Object Identifier no. 10.1109/TC.2009.92.

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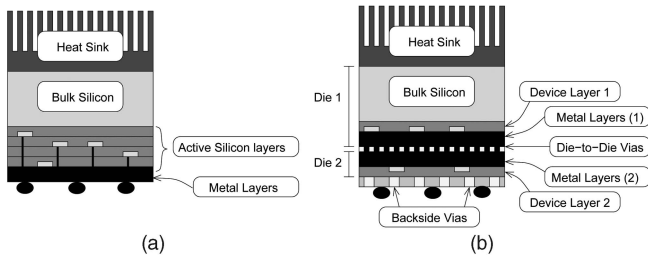


Fig. 1. (a) MLBS 3D technology. (b) Die bonding 3D technology.

3D technology processes the circuits using conventional planar fabrication processes and uses metal vias to bond the planar die vertically [24], [25]. Fig. 1b shows a die bonding 3D design using a 2-die stack. The die bonding approach involves depositing vias on the top metal layers of each of the two die and/or etching vias through the backside of the die, aligning the two die, and bonding them together. Under thermocompression, the vias fuse together providing both the die-to-die interconnects as well as the physical mechanism for holding the die together. After bonding, one die is thinned with chemical-mechanical polishing (CMP) down to only $\sim 10 \mu\text{m}$ allowing low impedance backside vias to be etched through, which provide input/output and power/ground connections.

There are many organizations for integrating multiple die in a 3D-integrated stack. Figs. 2a and 2b show 2-die face-to-face (F2F) and 4-die face-to-back (F2B) bonding topologies, respectively. The bonding process may be repeated to stack more than two die in any combination of face-to-face, face-to-back, and back-to-back organizations. The trade-offs between the different organizations have to do with the quality and pitch of the vias implementable at the different interfaces, and whether the vias interrupt the device layer. Fig. 2c shows a four-die stack that combines two F2F 2-die stacks with a B2B interface between the pairs of die. The alternating F2F topology, shown in Fig. 2c, is desirable

because the microarchitects and circuit designers can make use of the denser F2F interconnect at half of the die-to-die interfaces.

2.1 Physical Characteristics of Die-to-Die (d2d) Vias

The pitch and latency of the d2d via dictate the granularity at which a microprocessor circuit can be partitioned across the different die. Current academic implementations of 3D processes support d2d via pitches from 3 to $10 \mu\text{m}$ depending on the technology [26]. The embedded industry is already using $2.4 \mu\text{m}$ d2d vias in 3D structures [10]. As alignment technologies continue to improve, via pitches denser than $1 \mu\text{m}$ may soon be practical. As a point of reference, a recent study on implementing 3D caches pointed out that a state-of-the-art 6T SRAM cell takes up $0.7 \mu\text{m}^2$ (in 65 nm), which provides a d2d via density on the order of two d2d vias for every three SRAM cells [12].

The signal propagation delay between the two die is drastically reduced due to the reduced interconnect RC characteristics of the d2d vias. A d2d via is much smaller than the planar interconnect it replaces, and thus, reduces both resistance and capacitance. The thinning of the die also reduces the distance that a d2d via must cross to connect the two die. As mentioned before, the individual die are thinned to $\sim 10 \mu\text{m}$, and in an F2F organization, the d2d vias only need to cross the distance separating the two top metal layers. Depending on the technology, the d2d via height may be < 5 to $\sim 20 \mu\text{m}$ [26]. The delay to drive a signal from one die to another is less than one fan-out-of-four (FO4), which makes the cost of cross-die communication similar to a short length of traditional metal [11]. The B2B vias require etching through the bulk silicon substrate, and as a result, require larger structures in the current technology. The embedded processors industry is already using B2B vias with a $6 \mu\text{m}$ pitch and has reported that the next technology generation will support $< 4 \mu\text{m}$ B2B vias [10].

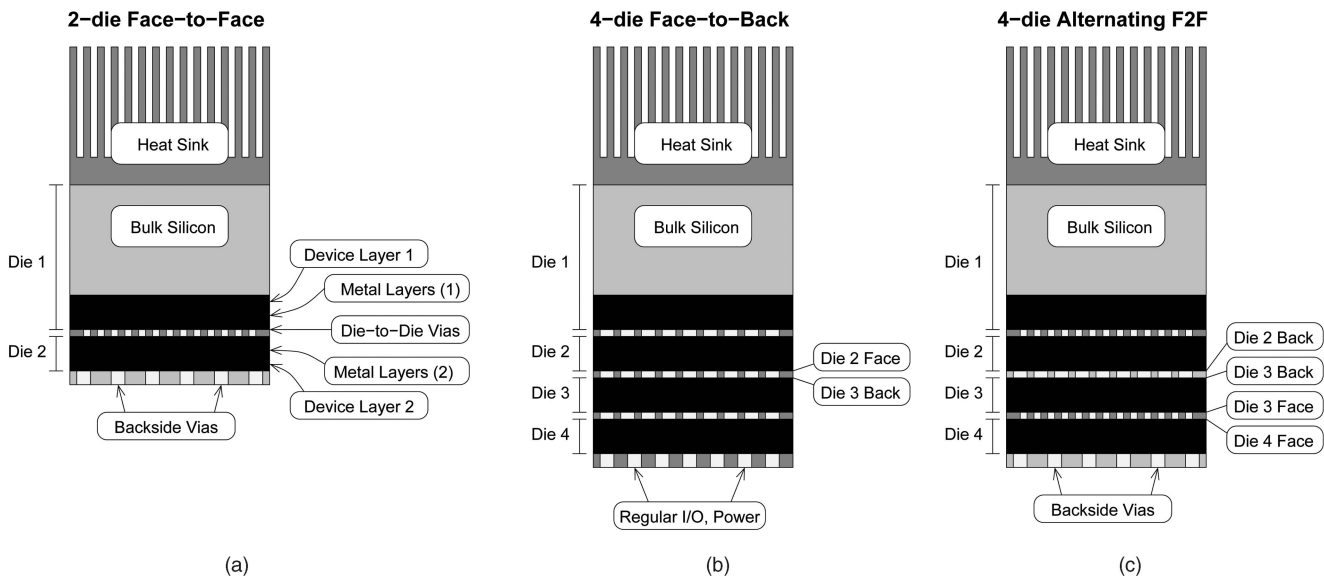


Fig. 2. (a) A 2-die 3D-integrated stack with face-to-face bonding, (b) a 4-die 3D-integrated stack with face-to-back bonding, and (c) a 4-die alternating face-to-face 3D IC (figures not drawn to scale).

In the rest of the paper, we assume an F2F interface for our 2-die stacked 3D-integrated circuits and an alternating F2F interface for our 4-die stacked 3D-integrated circuits.

3 PLANAR SRAM COMPONENTS

In this paper, we consider static random access memory (SRAM) arrays, which are regular-structured and wire-dominated circuits that show great promise for 3D implementations [6], [11], [12]. Many structures on the processor are variations of conventional SRAM arrays. These modules include caches, register files, branch predictor history tables, branch target buffers, physical and architected register files, reorder buffers, and payload RAM portions of the instruction issue logic. Different designs differ in the capacity (number of entries and bits per entry), timing constraints, and bandwidth requirements for reading and writing the array. On-chip cache memories are typically large SRAM array structures that occupy a majority portion of the processor die area. Caches reduce off-chip memory accesses by exploiting the concepts of temporal and spatial locality. Caches have a large capacity and require both tag and data arrays. Due to their large capacities, caches tend to be organized as banks in order to increase bandwidth while consuming less power. Caches are often subbanked to save power by sharing the sense amplifier circuitry among the subbanks. The basic cache design parameters are cache size, block size, and associativity. Register files are structurally similar to on-chip caches in that both consist of regular arrays of 6T SRAM cells. Register files have lower capacity requirements and do not have a tag array. But they are typically multiported with multiple read ports and multiple write ports in order to satisfy the bandwidth requirements of the data processing core. The register files are critical components of modern processors in terms of impact on clock frequency and instructions per second (IPC) rates. Other table-like structures such as the decoder PLAs, the microcode ROM in x86 processors, and the lookup table for SRT dividers may also observe similar benefits from a 3D implementation.

The planar SRAM array-based components have a regular array of memory cells that are easy to partition across multiple die. The main SRAM array can be viewed as a set of wordlines running horizontally and a set of bitlines running vertically. A row decoder drives the wordlines, which control the access transistors of the data storage cells. The bitlines are read by sense amplifiers at the bottom of the array. In the case of tagged SRAM structures such as caches, a similar array stores the tags and a set of comparators performs the tag match operation.

We describe four existing techniques to increase the performance and reduce the power consumption in SRAM arrays, namely banking, subbanking, hierarchical wordlines (HWLs), [27] and multiporting.

3.1 Memory Banking Technique

Memory banking is a technique that divides the memory array into multiple submodules called banks. By dividing the array into multiple banks and accessing only the bank that contains the required data, significant power saving can be realized. Memory banks can also be used to enhance

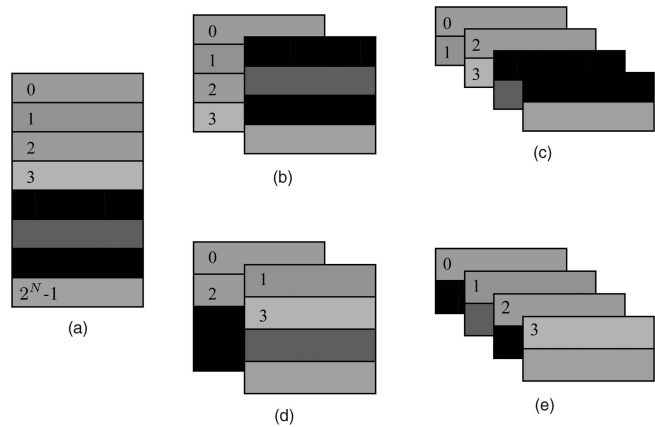


Fig. 3. Memory banking technique: (a) one single array, (b) two banks with higher order interleaving, (c) four banks with higher order interleaving, (d) two banks with lower order interleaving, and (e) four banks with lower order interleaving.

bandwidth, while not requiring a significant wire area increase as in the case of multiple ports. Fig. 3a shows a single memory array with addresses ranging from 0 to $2^N - 1$. Fig. 3b shows the same array divided into two banks using higher order address interleaving technique. Higher order address interleaving divides the memory array into banks based on the higher order address bits. If the array contains 2^N locations (0 to $2^N - 1$), then one bank contains addresses from 0 to $(2^{N-1} - 1)$ and the other bank contains addresses from 2^{N-1} to $2^N - 1$. Fig. 3c shows the array divided into four banks using higher order interleaving. Figs. 3d and 3e show the memory array divided into two and four banks based on lower order address interleaving. Lower order address interleaving uses the lower order address bits to identify the banks. Hence, this technique divides the odd and even addresses into separate banks. If the requested data reside in only one of the multiple banks, the other banks need not be accessed and do not consume dynamic power.

Banking can also be used to enhance the bandwidth. If the requested data values reside in different banks, we can simultaneously obtain values out of multiple banks, thus, mimicking the effect of a multiported memory array. Note that it is possible for two or more simultaneously required data to reside in the same bank. When multiple addresses target the same bank, we have a bank conflict. In case of bank conflicts, we need a buffer mechanism that stores and reissues the requests so that the target bank provides the requested data values in later clock cycles.

3.2 Memory Subbanking Technique

Memory subbanking is a technique used to save power in the bitline and sense-amplifier network [28]. A cache block is divided into a number of subbanks. A part of the block offset bits is used to select the subbank that contains the required data. Since data are read out from only one subbank at a time, a common set of sense amplifiers can be shared across the subbanks, cutting down on the cache power. Fig. 4a shows a data array of a cache with each cache line consisting of four data words. In the absence of subbanking, all four data words are read out and sensed using sense amplifiers.

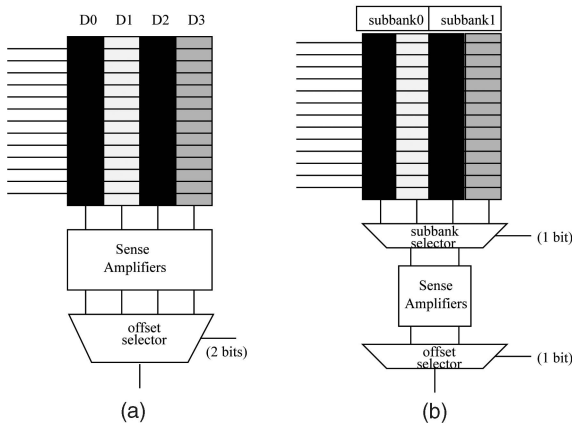


Fig. 4. Memory subbanking technique: (a) data array with each line consisting of four data words, (b) data array divided into two subbanks with two data words per subbank.

The required word is then chosen using the offset bits in the address. Fig. 4b shows the same data array divided into two subbanks, with each subbank consisting of two data words. In the case of subbanking, the selector logic selects between the two subbanks and feeds the data from only one subbank into the sense-amplifier circuitry. Hence, the subbanks can share the sense-amplifier circuitry and save significant amount of power. Subbanking also helps in saving the bitline precharge power, since only the selected subbank needs to be precharged.

3.3 Hierarchical Wordline Technique

Wordlines in SRAM arrays are heavily loaded by the access transistors, which are two per SRAM cell across the whole rows of SRAM cells. Hence, wordlines contribute significantly to the overall delay of SRAM access. Hierarchical wordline technique improves performance by reducing the latency of driving a wordline. An HWL structure uses a global wordline (GWL) to drive multiple shorter subwordlines distributed across the SRAM array. With an HWL structure, the wordline loading is reduced, thus, enhancing the overall performance. As an unintended side effect, the HWL technique also worsens the wire complexity of the wordlines since the wiring requirement of wordlines is essentially doubled.

3.4 Multiporting Technique

Modern superscalar processors capable of issuing many instructions per cycle require a large number of read and write ports from the register file, and the size of an SRAM cell increases dramatically with increasing port requirements. Fig. 5 shows SRAM cells with (a) one port, (b) two ports, and (c) four ports. While Fig. 5 illustrates a simplistic SRAM cell design (for example read-port isolation transistors have been omitted for clarity), it demonstrates the quadratic increase in area with respect to the port count. The area explosion forces the wordlines and bitlines to increase in length that, in turn, increases both access latency and power consumption. Many microarchitecture-level proposals have been made to deal with the size, latency,

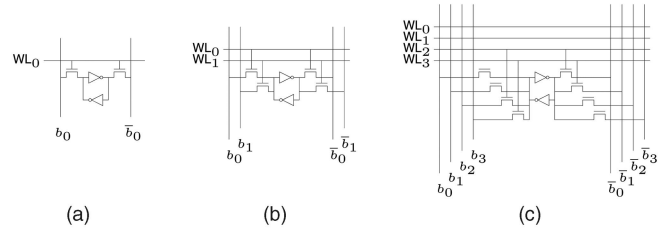


Fig. 5. Quadratic area increase of a simple SRAM bit cell as the number of ports increases from (a) one to (b) two to (c) four ports.

and power of the register files, including register caching [29] and register file banking [30]. While these techniques can reduce the average latency of register file access, they significantly complicate the processor data and control paths. Increases in processor clock frequency and the relative decrease of the speed of wires only exacerbate the problem. Another example of the poor scaling of register file latency and area can be found in the Alpha 21264's 4-issue integer execution core, which would normally require an 8-read port, 4-write port register file. Instead, the designers chose to duplicate the entire content of the register file such that each copy only needs half as many read ports [31]. Two full copies of a moderately ported register file proved to be smaller and faster than a single highly ported structure. Since the register file is dominated by wire, 3D may provide an effective means for controlling the latency, power, and area of the large register files required by modern high-performance processors.

Now, we propose 3D-integrated designs of the SRAM array structures. For illustration purposes of the planar baseline and our 3D-SRAM designs, we consider caches and register files as respective examples of banked and multiported SRAM array designs.

4 3D IMPLEMENTATIONS OF BANKED SRAM ARRAYS

Long metal wires used to route global signals in banked SRAM arrays such as caches suffer from large RC delays and power consumption. The address wires that route from the edge of the SRAM array to a bank are examples of such wires. Fig. 6a shows an 8-bank SRAM array with the critical wire path from the bottom-left corner of the cache to the farthest bank. One option for a 3D-integrated SRAM array design is to stack banks on top of each other. Another option is to split the arrays on multiple layers. We describe the design of SRAM arrays in 3D based on bank stacking and array splitting.

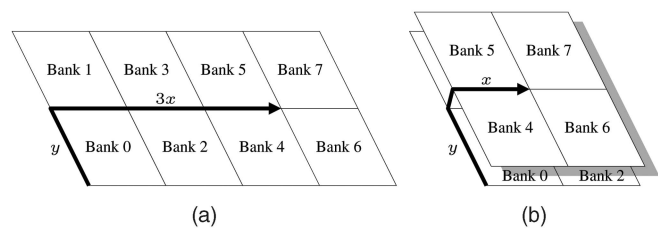


Fig. 6. (a) A planar 8-banked array showing the worst-case distance to the farthest bank and (b) a 3D bank-stacked SRAM organization.

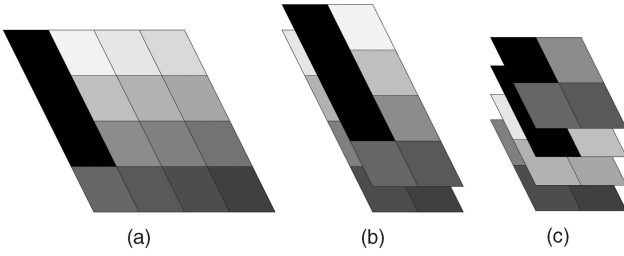


Fig. 7. (a) A banked planar array with 16 banks, (b) a 2-die stacked 3D array with left-to-right stacking, (c) a 4-die stacked 3D array with top-to-bottom stacking across the F2F interfaces and left-to-right stacking across the B2B interface.

4.1 Bank Stacking 3D

Fig. 6b shows a 3D bank-stacked organization with a 2-die stack. A 3D bank-stacked design was previously proposed by Reed et al. [6]. There are two possible orientations for bank stacking. The banks can be stacked left-to-right, as shown in Fig. 6b, or top-to-bottom. Fig. 6b shows how the bank stacking results in a 67 percent reduction ($3x \rightarrow x$, where x is the bank width) in the horizontal component of the wiring to and from the banks. Because the stacking of the banks occurs only in one direction, the vertical component of the bank wiring is unaffected, thus, reducing the wire length savings to 50 percent (assuming that $x = y$, where y is the bank height). Note that the actual reduction may be slightly less than 50 percent due to the small overhead incurred by the interdie via in the z-dimension. Overall, the reduction in wire length translates into a reduction of power and delay.

When we have more than two die on the 3D-integrated stack, we can combine the bank splitting strategies to increase the benefits of 3D. The fastest bank-stacked 3D SRAM organizations may use different bank partitioning across the stacked layers. The application of a particular bank stacking technique reduces the latency of a critical wire by a significant amount such that it may no longer be the worst delay in the circuit. A different style of partitioning can then address the new worst delay. With a 4-die stack, we can stack the banks top-to-bottom across the F2F interfaces and left-to-right across the B2B interface. Fig. 7a shows a 16-banked planar SRAM array design, that is stacked left-to-right on a 2-die stack as shown in Fig. 7b. With a 4-die stack option, the next partitioning happens top-to-bottom, thus, producing a 4-die footprint that is more evenly balanced in both horizontal and vertical dimensions.

4.2 Array Splitting 3D

We evaluate partitioning individual rows and columns of the SRAM arrays within a bank and stacking them upon themselves. Array splitting can reduce the lengths of either the wordlines or the bitlines depending on the orientation of the split. Fig. 8a shows the details of a small SRAM data array. With bank stacking, the decision for a horizontal or vertical split largely depends on which dimension has more wire delay. At the array level, the orientation of splitting has a greater impact on the implementation of the array itself.

The first array-split configuration stacks columns on columns. Fig. 8b illustrates the column-stacked 3D array. In this design, the single long wordline has been replaced by a pair of parallel wordlines. The row decoder must drive the

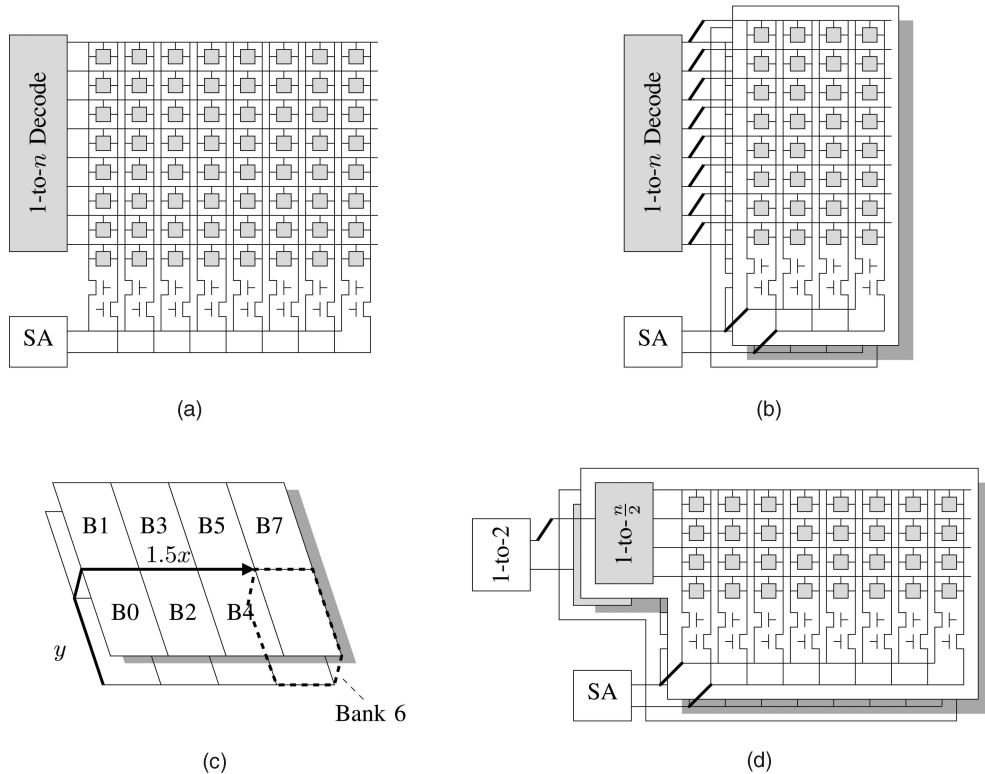


Fig. 8. (a) A planar SRAM array, (b) the same array in 3D with column-on-column stacking, (c) the bank-level organization using column-stacked arrays, and (d) a 3D array using row-on-row stacking.

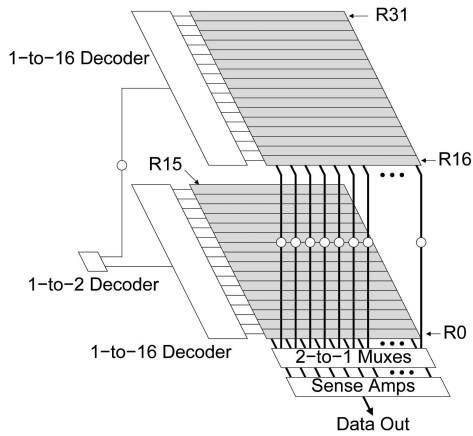


Fig. 9. Register partitioning 3D register file. A $\circ$ represents a die-to-die via.

wordlines on both of the die. This requires one d2d via per wordline. At the bottom of the array, the column select multiplexers have been split across the two die, thus, requiring additional die-to-die vias. This organization reduces latency and power due to reduced wordline lengths. As shown in Fig. 8c, these benefits are in addition to reductions in the bank-level wire lengths.

The second array-split organization stacks rows on rows. Fig. 8d shows a row-stacked 3D array. The row-stacked array requires the row decoder to be partitioned across the two die. We first decompose the 1-to- n decoder into a 1-to-2 decoder followed by two 1-to- $\frac{n}{2}$ decoders. Even though the two 1-to- $\frac{n}{2}$ decoders are stacked on top of each other, the 1-to-2 decoder will only activate one of them that avoids stacking of thermally active components. The lengths and loading of the wordlines remain the same as in the planar organization, but the lengths of the bitlines reduce to half. Similar to the column-stacked organization (Fig. 8b), there are latency and power benefits due to wire reduction at both the array and bank levels.

5 3D IMPLEMENTATIONS OF MULTIPORTED SRAM ARRAYS

We propose 3D implementations of multiported SRAM arrays such as register files. There are many possible designs for multiported SRAM arrays in 3D integration technology. We propose four different strategies for partitioning the multiported SRAM arrays across multiple die.

5.1 Register Partitioning (RP)

A register-partitioned (RP) 3D SRAM array with a 2-die stack splits half of the entries and places them on the vertically stacked die. Fig. 9 illustrates a 32-entry register file implemented on a 2-die stack, where the bottom die contains registers R0-R15 and the top die contains R16-R31. A result of this topology is that the vertical distance (along the bitlines) has been halved, which can greatly reduce the latency and power associated with toggling the bitlines. The row decoder height has also been halved, which reduces the length of the critical path associated with accessing the farthest entry in the register file. The overall

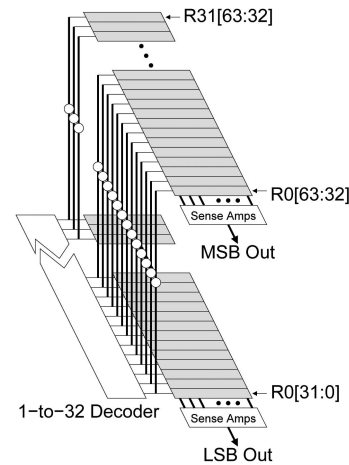


Fig. 10. Bit partitioning 3D register file. A $\circ$ represents a die-to-die via.

footprint of the register file has also been halved, which may enable more compact processor floorplans.

To implement a 4-die RP 3D register file, the register entries would be partitioned such that one quarter of the entries resides on each die. The row decoder can be further decomposed in a manner similar to the 2-die stack.

Note that while Fig. 9 shows a 32-entry register file with a single read port, the register file of modern high-performance processors may have 80 [31] or 128 [32] entries. As the number of in-flight instructions increases, the register file size also grows accordingly. The large number of read and write ports also exacerbates the area and wire lengths of the register file. Furthermore, the width of each register has increased over time from 32 to 64 bits, and even 128 bits for some instruction set architectures (e.g., Intel's SSE3 ISA extension provides 128-bit registers).

5.2 Bit Partitioning (BP)

The bit-partitioned (BP) 3D register file stacks higher order and lower order bits of the same register across different die. The BP register file can be viewed as the dual of the RP organization: RP folds the register file upon itself in the horizontal direction while BP folds in the vertical direction. Fig. 10 shows a 2-die 64-bit 32-entry bit-partitioned register file, where the bottom die stores the least significant bits of the register values and the top die stores the most significant bits. The bit-partitioned register file reduces the wire length and gate loading on the wordline, which provides both latency and energy reduction benefits.

While Fig. 10 shows the bits of each register partitioned by significance, one could instead store the bits in odd positions on one die and the bits in even positions on the other die. Choosing one over the other does not impact the area, latency, or power of the 3D BP register file. However, the choice should be made to match the data paths throughout the rest of the processor. For example, if one implements a 3D integer arithmetic unit partitioned by significance ($X[0:31] + Y[0:31]$ on one die, $X[32:63] + Y[32:63]$ on the second die) [13], then the register file bit partitioning should also be arranged by significance to avoid unnecessary d2d routing between the outputs of the register file and the inputs of the arithmetic units.

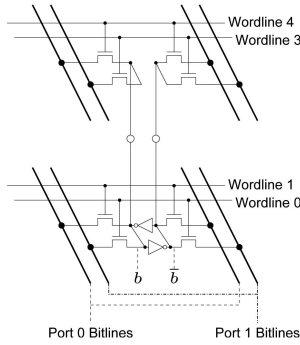


Fig. 11. Port-split 3D register file with two vias per bit cell. A $\circ$ represents a die-to-die via.

The BP 3D register file requires the row decoder outputs to be fanned out to the different die. This extra communication incurs a small overhead, but the latency reduction due to the halving of the wordline length provides a significant net benefit.

For a 4-die organization, the overhead of row decoder output fan-out increases, but only slightly because the fan-out trees scale well with increasing die count.

5.3 Port Splitting (PS)

The individual SRAM cells of caches are designed to be very small so as to increase the capacity of the cache, while the area-per-bit for a register file cell is dominated by the wordlines and bitlines for implementing multiple read and write ports. Tsai et al. [12] suggested that the relative size of a 6T SRAM cell and a d2d via makes it difficult to implement an individual 6T cell across multiple die. However, register file SRAM cells have a substantially larger footprint (due to the high port count), which may provide the opportunity to allocate one or two d2d vias for each cell. Fig. 11 shows a 2-die port-split (PS) SRAM cell, where each die contains the bitlines, wordlines, and access transistors for half of the ports (either read or write). Two d2d vias are required per bit cell to route the outputs of the chained inverters to the second die.

The PS register file provides substantial benefits in terms of area footprint reduction. Stacking the wordlines on top of each other halves the height and stacking the bitlines halves the width. A 50 percent reduction in both dimensions leads to an overall footprint reduction of ~ 75 percent for the SRAM array. The total footprint reduction may be slightly less than 75 percent because structures like the row decoder and sense amps may not observe as large of a compaction benefit. This substantial area reduction translates into latency and energy savings because both bitline and wordline lengths have been halved.

Depending on the multiported SRAM design parameters and the relative sizes of d2d vias, it may or may not be possible to allocate two d2d vias per bit cell. Fig. 12 shows an alternative implementation of a 2-die PS 3D register file cell, where only a single d2d via is used to route the data bit b to the second die. On the upper die, an extra inverter recomputes the complement bit $\bar{b}$. This shows that logic duplication may be used to trade-off against excessive interdie communication.

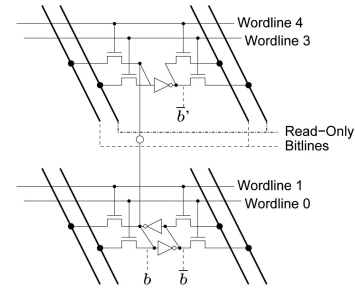


Fig. 12. Alternate port-split 3D register file that only uses one via per bit cell. A $\circ$ represents a die-to-die via.

A limitation of the single-via configuration is that the ports on the top die may not be able to optimally support write operations because there is no path to access the “true” $\bar{b}$ storage node.¹ This limitation is likely not critical as the number of write ports is typically much less than the number of read ports.

With two d2d vias per cell, another design alternative would be to split the back-to-back inverters across the two die. This would place all of the b bitlines on (say) the bottom die and all of the $\bar{b}$ bitlines on the top cell. We do not evaluate this configuration as it has several disadvantages. First, the wordlines must be replicated across both die (similar to the BP register file configuration), which eliminates the wirelength reduction in one dimension. Second, splitting the differential bitlines across more than one die may require designing sense amplifiers that are themselves partitioned across more than one die.

5.4 Hybrid Configurations for Larger 3D-Integrated Stacks

There are multiple ways to partition the SRAM arrays based on a combination of the previously described implementations of the 3D SRAM arrays. In this section, we describe some of the examples of hybrid strategies.

Hybrid SRAM arrays implemented across more than two die use a combination of the partitioning strategies. This may be particularly useful in an alternating F2F/B2B die-stacking organization, where the available d2d via density changes between pairs of die. In a 4-die stack with alternating F2F interfaces, one could first use register partitioning to assign half of the registers to die 0/1 and the other half to die 2/3, which limits the usage of the coarser B2B vias to the periphery of the main SRAM array. Then between each pair of F2F die, port splitting could be employed to exploit the denser F2F interface within the SRAM array.

We illustrate the hybrid partitioning with a branch prediction table design. Control flow predictors such as branch predictors constitute a significant portion of the front end in modern processors. In particular, we consider a branch direction predictor table based on two-bit saturating counters as shown in Fig. 13a. The branch direction predictor table shown in Fig. 13a is an SRAM array structure that consists of a column of least significant bits

1. One could conceivably build a single-ended write port that only changes the value of node b and relies on the SRAM cell itself to override the $\bar{b}$ node, but this would increase the write latency and substantially increase the duration of the short-circuit interval, where both PMOS pull-up and NMOS pull-down circuits are active.

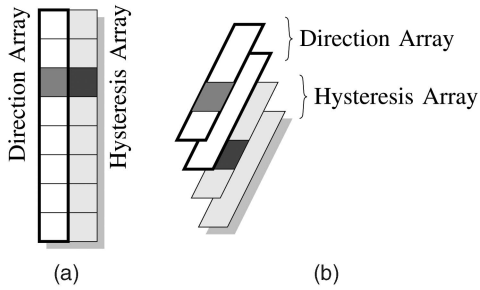


Fig. 13. (a) Planar branch direction predictor array. (b) 3D branch predictor array partitioned into two separate 3D subtables.

called the hysteresis bits and a column of most significant bits called the direction bits. The processor needs the direction bit for making the initial prediction as well as during the update/training phase, while the hysteresis bit is only needed for the update. For such branch direction predictors, we can first partition the counters into two separate arrays: one array to store the direction bit (msb) and the other to store the hysteresis bit (lsb) [33]. We implement the two arrays by partitioning them across two die each, as shown in Fig. 13b. Such a partitioning may not only result in more evenly balanced footprints but also other benefits such as placement of the more frequently accessed arrays on the die closer to the heat sink, resulting in latency, power, and thermal advantages.

6 RESULTS

We simulate the critical path circuitry of the SRAM arrays with varying design parameters such as array capacity and planar versus 3D organization. To optimize our planar and 3D designs, we sweep through a range of transistor sizings. We use the configurations that minimize overall access time.

We use HSPICE to obtain the critical path latency and the power consumption of the entire SRAM array. Our HSpice simulations use 65 nm predictive technology models [34]. We used 130 nm wire parameters from Intel and extrapolated down to 65 nm [35]. Our 3D circuit designs use d2d via pitches of 1 μm for the face-to-face and 2 μm for backside interfaces. While the pitch of current manufacturable d2d vias is already only 2.4 and 4 μm for F2F and B2B interfaces [10], respectively, we assume that the d2d sizes will continue to scale at least for a few generations. Furthermore, we assume a copper-to-copper 3D integration technology, which means that the d2d vias are made of the same copper as the traditional on-die interconnects, and therefore, have similar unit resistance and capacitance. The distance between the top metal layers on the two die is very small [25], [36] and the pitch of the die-to-die vias is on the same order as the top level metal [37]. We model the die-to-die interconnect as 10 μm of top-level metal plus the corresponding via contact resistance based on the published data [25], [36], [37]. For our interconnects, we use a distributed RC network model [38]. We calculate the lengths of wires between various subblocks based on the critical path analysis and combine the length data with other physical parameters [34], [35] to obtain the resistance and capacitance

TABLE 1
The Impact of 3D Technology on SRAM Array Latency (2-Die F2F, 4-Die Alternating F2F, All 3D Circuit Latencies Reported Are for Array-Split Configurations)

Cache (KB)	Latency (% Benefit), ps			Cycles (3 GHz)		
	Planar	2-die 3D	4-die 3D	Planar	2-die 3D	4-die 3D
16	458	449 (2)	450 (2)	2	2	2
32	752	635 (16)	584 (22)	3	2	2
64	1232	885 (28)	731 (41)	4	3	3
128	1716	1381 (20)	1233 (28)	6	5	4
256	2732	1929 (29)	1513 (45)	9	6	5
512	3663	2864 (22)	2461 (33)	11	9	8
1024	5647	3945 (30)	3066 (46)	17	12	10
2048	7456	5774 (23)	4875 (35)	23	18	15
4096	11415	7854 (31)	6096 (47)	35	24	19

data for the various interconnects. We model repeaters on global wires to alleviate the quadratic delay increases.

6.1 Bank and Array-Stacked 3D SRAM Benefits

We simulate a cache read operation with varying design parameters such as bank organization, array capacity, line size, and planar versus 3D organization, while including the bank- and array-level circuit details. The simulations account for all of the wires and drivers to get from the array boundary to the farthest bank. The request propagates through the decoder tree, the wordline, the SRAM cell, the bitlines, the column multiplexor, the sense amplifier, and the way multiplexor. This signal is then driven back through a series of wires and bank multiplexors to finally return to the edge of the array. While the critical timing path involves reading a single SRAM entry, our power simulations include the activity associated with enabling *all* of the SRAM cells and their bitlines in the selected array row. We perform a similar analysis for the critical path through the tag array, including the tag comparators and the interaction with the way-select multiplexors on the data side. We use the worst-case timing path through either array.

We evaluate our 3D array designs against conventional planar arrays. We consider array sizes varying from small 16 KB level-one (L1) caches up to large 4 MB last-level (L2/L3) caches. Our baselines for comparison are planar caches with transistor sizings and bank configurations chosen to minimize latency.

Table 1 reports the size and latency of the baseline planar arrays, the latency of the 3D array-split circuits, the relative latency reduction, and the latency in terms of 3 GHz clock cycles. Note that we observed the array-split configurations to provide additional latency reductions as compared to the bank-stacked configurations. In the rest of the paper, we provide results for the most latency saving 3D configuration, which is the array-split configuration, unless we state otherwise. The observed latency reduction benefit varies by array size. The 3D organization provides more benefit to the larger arrays because these structures have substantially longer global wires to route signals between the array edge and the farthest bank. The latency improvement in Table 1 does not increase monotonically because the best configuration between the 2D and 3D arrays may involve a different number of banks, which, in turn, changes the

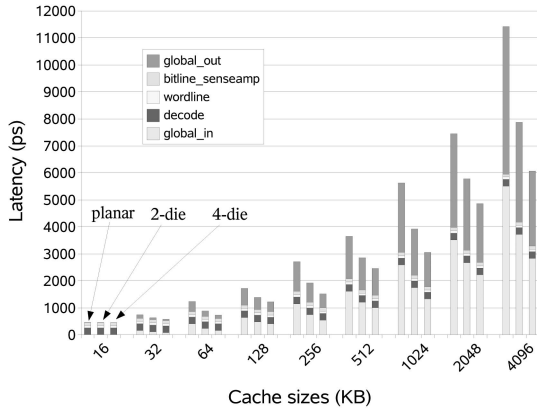


Fig. 14. The componentwise breakdown of the array latencies reported in Table 1.

relative benefit. For many 3D configurations, the latency reduction is sufficient to reduce the overall number of clock cycles for the access as shown in the last three columns of Table 1. For the smaller arrays, the wire delays comprise a smaller relative fraction of the overall latency, and therefore, the effect of reducing these delays is less.

To understand where 3D integration has the greatest benefit, Fig. 14 illustrates the latency contribution of the different components that comprise an SRAM array. Depending on the array size, different 3D topologies may be required to provide the best latency improvements. As an example, the bank-level routing latency for the 2 MB planar cache makes up over 55 percent of the total latency. Correspondingly, the fastest 3D 2 MB organization exhibits the greatest latency reduction in these timing components, as shown in Fig. 14. On the other hand, moderate-sized (64-512 KB) cache delays are not as dominated by the global routing. In these instances, a 3D organization that targets the intra-SRAM delays provides more benefit. Fig. 14 shows that, in particular, the delay reduction associated with toggling and sensing the bitlines accounts for a substantial amount of the overall benefit, although other timing components also observe some improvement. In general, circuit paths dominated by long wire RC delays have the greatest potential for improvement from a 3D reorganization. In contrast, the row decoder consists primarily of logic, and the results in Fig. 14 are consistent with the expectation that 3D would not provide much benefit for this component of the overall cache latency.

The 3D organization of the cache structure reduces critical wire lengths, which reduces the total wire capacitance providing both performance improvements *and* energy reductions. For the same cache configurations discussed in the previous section, Table 2 shows the energy consumed per read access for both the planar and 3D configurations. The overall savings range from 2-18 percent, varying due to the differences in the optimal banking configurations, transistor sizings, and SRAM aspect ratios. Note that the energy benefits are in addition to the latency reduction benefits as reported in Table 1. Thus, we show that 3D-integrated SRAM array designs can provide simultaneous latency and energy reduction benefits as compared to the corresponding planar designs.

TABLE 2
The Impact of 3D Technology on SRAM Array Energy for a Cache Read Operation (All Reported 3D Energy Numbers Are for Array-Split Configurations)

Cache (KB)	Energy, nJ			Percent	
	Planar	2-die 3D	4-die 3D	2-die 3D	4-die 3D
16	76.93	74.12	75.37	4	2
32	86.37	86.90	84.06	-1	3
64	87.92	86.49	83.69	2	5
128	102.35	99.17	100.30	3	2
256	106.39	102.70	100.05	3	6
512	108.64	106.38	105.55	2	3
1024	120.75	114.10	107.28	6	11
2048	129.41	121.94	117.86	6	9
4096	150.62	133.32	123.78	11	18

Fig. 15 shows the energy consumption distribution of the array read operation. A component's relative contribution to total latency does not always reflect its impact on energy. The row decoder logic accounts for a considerable amount of latency, but it only needs to switch a single wordline and so does not consume much energy. On the other hand, a single bitline switch and the corresponding sense amplifier only incur a modest energy cost, but this cost must be multiplied by the number of bits read and by the set associativity, which adds up to account for a substantial portion of total access energy. The 3D organization reduces the energy in the most wire-dominated portions of the arrays, namely the bank-level routing and the bitlines.

Fig. 16 shows the impact of HWL technique on the benefits of 3D caches. Each bar in Fig. 16 shows the percent latency reduction benefit of the 3D design compared to the corresponding planar baseline design. For example, the first bar shows the latency reduction benefit of a 2-die 3D cache with no HWL, as compared to a planar design with no HWL. The second bar shows the latency reduction benefit of a 2-die 3D cache with HWL, as compared to a planar design with HWL. For every cache size, 3D benefits are even larger for the design with HWLs. Since HWLs enhance performance, 3D-integrated SRAM designs are even more attractive in high-performance planar designs that use techniques such as HWL to improve the performance.

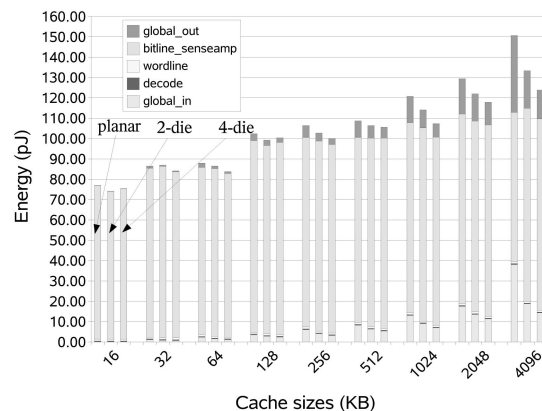


Fig. 15. The componentwise breakdown of the SRAM array energy reported in Table 2.

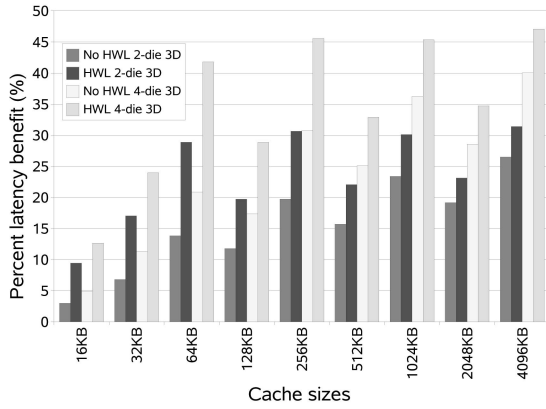


Fig. 16. The effect of HWLs on 3D benefits.

Fig. 17 shows that increasing the line size reduces the benefit of 3D caches. For smaller cache sizes (from 32 to 512 KB), as the line size increases, 3D latency benefits decrease. For smaller cache sizes, global wire delays are already a small fraction of the total delays (intra-array delays dominate total delays), and thus, provide less benefits from 3D implementations. When the line size increases, intra-array delays further increase, and the global wire delays further decrease, thus, reducing 3D latency benefits. For larger cache sizes, total delays are dominated by global wire delays and not the intra-array delays. Hence, as the line sizes increase, we do not observe as much reduction in latency reduction benefits.

6.2 Multiported 3D SRAM Benefits

We generate custom critical path netlists for different register file configurations based on parameters such as the number of entries, bit width per entry, number and type of ports, and planar versus 3D organizations. The 3D implementations of the register file substantially reduce both the latency and energy associated with long wires. We compare baseline planar implementations of register files to a variety of 3D organizations explained earlier in Section 5.

6.2.1 Effect of Increasing the Number of Entries

We evaluate register file designs in a high-performance superscalar processor. In particular, we consider a high-performance superscalar machine in which the processor

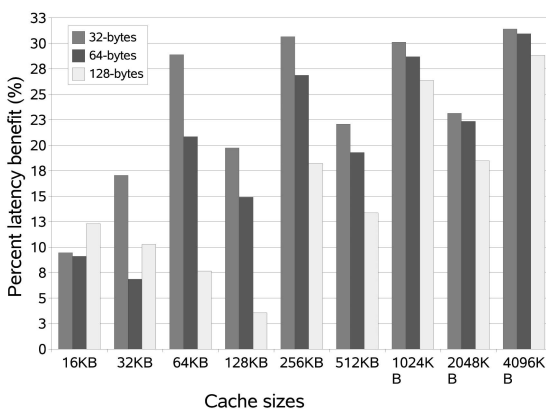


Fig. 17. The effect of increasing line size on 3D integration benefits.

TABLE 3
Benefits of 3D Multiported SRAM Configurations
with Increasing Number of Entries

RF entries	Planar (ps)	% 2-die			
		BP	RP	PS	
16	258	38	3	10	
32	307	34	4	12	
64	334	27	7	13	
96	383	25	5	11	
128	398	23	9	14	
160	442	24	14	19	
192	458	24	15	19	

RF entries	Planar (ps)	% 4-die					
		BP-BP	RP-RP	PS-PS	BP-RP	BP-PS	
16	258	43	4	17	40	43	
32	307	37	8	17	37	40	
64	334	32	12	20	38	43	
96	383	25	13	21	32	39	
128	398	26	14	22	33	36	
160	442	25	20	28	35	40	
192	458	27	21	28	35	58	

core can execute four instructions per cycle and can operate on 64-bit operands. Each of the instructions requires two read ports and one write port. Furthermore, to retire multiple instructions per cycle, additional read ports are needed to read the register contents before updating the architected (committed) register file. The total port requirement is 12 read ports and four write ports for a four-wide issue machine. We simulate register files with sizes ranging from 16 up to 192 entries to model the register file capacity demands of current and future processors. Table 3 tabulates the planar latencies of each of the register files and percent latency reduction for each of the 3D configurations considered. Table 3 shows that, for 2-die stacks, the BP design provides the largest latency reduction benefit when compared to the corresponding RP and PS designs. The wordline is heavily loaded by the two access transistors per bitline column, and so splitting the wordline across two dies reduces a major component of the wire latency. Since the BP design reduces the wordline loading, it provides a large latency reduction benefit. Note that, as the number of entries increases, latency reduction benefits of BP designs decrease. This is because, as the number of entries increases, the height of the overall structure increases, thus, making the row decoder and bitline/sense-amplifier delay to become increasingly critical. Bitlines become more heavily loaded, thus, reducing the contribution of the wordline load to the overall latency. Two-die stacked RP designs do not provide as much of a latency reduction benefit as the BP design. But, as the number of register entries increases, the benefits of the RP design also increases. Although the PS designs have a substantially smaller footprint, they do not provide the fastest performance due to the fact that the access transistor loading on the wordlines has not been reduced. With increasing number of entries, the wire complexity keeps increasing. Hence, 2-die stacked PS designs also exhibit increasing savings with increasing number of entries.

In the case of 4-die stacks, there are more design options when implementing 3D designs. The notation of X-Y for 4-die stacks means that the B2B interface has an X partitioning and the F2F interfaces have a Y partitioning.

TABLE 4

Access Latencies of Register Files for a 4-wide, 64-bit Superscalar Processor, and the Percentage Benefit Compared to the Baseline Planar Implementation, for Increasing Entries

RF entries	Planar (ps)	2-die 3D (ps)	4-die 3D (ps)	% 2-die	% 4-die
16	258	159	146	38	43
32	307	202	185	34	40
64	334	242	190	27	43
96	383	286	234	25	39
128	398	306	253	23	36
160	442	337	265	24	40
192	458	349	193	24	58

An RP-BP organization places one half of the registers on die 0/1 and the other half on die 2/3 (RP across the B2B interface), and then within each pair of die, each register's bits are split half on one die and half on the other. With 4-die stacked multiported SRAM array designs, the BP-BP configuration saves more latency than either of the RP-RP and PS-PS designs. However, with increasing number of entries, the RP-RP and PS-PS designs provide comparable benefits as the BP-BP designs due to higher load reduction and wire reduction, respectively. Of particular interest are the hybrid configurations, BP-RP, and BP-PS, which consistently provide large benefits as compared to the plain 4-die stacked configurations. This makes sense since one 3D partitioning technique may reduce the latency of a critical wire delay by a significant amount such that it is no longer the worst delay in the circuit. A second different 3D partitioning can then address the new worst delay. In fact, the maximum benefit for each register file size among all the 3D designs is given by the BP-PS design. The PS configuration is via-intensive, and thus, an unsuitable candidate for the B2B interface that requires vias etched through silicon. However, PS reduces wires in both horizontal and vertical dimensions resulting in higher wire savings and can be implemented beneficially across the F2F interfaces. This latency reduction may potentially be converted into a clock frequency increase or a reduction in the number of cycles necessary to access the register file.

The latency reduction benefits of the hybrid configurations highlight the generality of the 3D stacking techniques. For different processor configurations, the critical wire delay components within the register file will likely be different. The benefits of 3D integration are not limited to specific processor microarchitectural parameters; that is, for each situation, a circuit designer can choose the appropriate 3D stacking strategy to provide a latency reduction.

Table 4 shows the best absolute latencies of planar, 2-die stacked 3D, and 4-die stacked 3D register files and the corresponding percentage benefits. From the table, we can see that the 2-die stacked designs provide 23-38 percent latency reduction benefit over planar while 4-die stacked designs provide 36-58 percent latency reduction benefit. From the data, we conclude that a 4-die stacked design may not be able to provide 2X benefit as compared to the 2-die stacked design. The latency reduction benefit is not doubled for a 4-die stack as compared to a 2-die stack because the

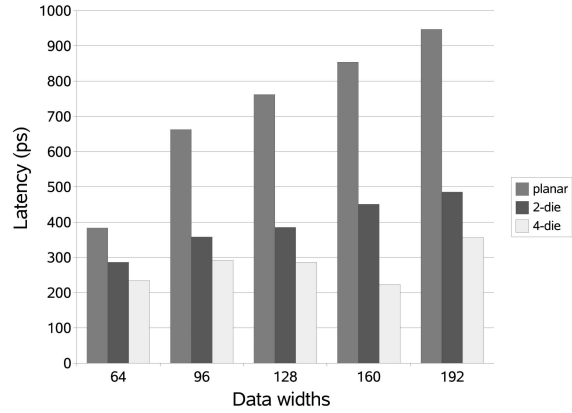


Fig. 18. Access latencies of a 96-entry register file for a 4-wide superscalar processor for increasing data widths.

4-die stacked designs require B2B vias to go through silicon, which adds area overhead and reduces the latency benefits.

6.2.2 Effect of Increasing the Data Widths

In some microarchitectures, the physical register file also contains some instruction status such as instruction pointer information, which is required for maintaining in-order retirement of instructions; each of the registers may contain as many as 160 bits of data [39]. We explore a 96-entry register file with increasing data widths from 64 to 192-bits in increments of 32-bits and report the benefits in Fig. 18. From the figure, we can deduce that the 2-die stacked designs provide 25-50 percent latency reduction benefit over planar while the 4-die stacked designs provide 40-75 percent latency reduction benefit. Note that the latency savings are higher in this case as compared to increasing number of entries since increasing data widths aggravate the already heavily loaded wordlines. Note that the planar register files and the 2-die stacked 3D register files exhibit a monotonously increasing latency behavior. On the other hand, 4-die stacked 3D register files have a nonmonotonous behavior due to the fact that the splits across the F2F and the backside interfaces have different overheads associated with them. Hence, even though a certain partitioning may be the most beneficial in terms of wire reduction (e.g., port split), the interdie via overhead may alter the overall benefits.

6.2.3 Effect of Increasing Port Requirements

We explore a 96-entry, 64-bit-wide register file with increasing issue widths from 2 to 6-way as shown in Fig. 19. Since every instruction requires at least two read ports to read the source operand registers and one write port to write the destination register, each increment in issue width adds two additional read ports and one additional write port to the register file. Note that the planar latency degrades at a much faster rate than the 3D latencies. With increasing issue widths, the latency reduction benefits of 3D designs continuously increase compared to the corresponding planar designs.

6.2.4 Energy Benefits of Multiported 3D SRAMs

In addition to reducing the access latency of the multiported SRAMs, the 3D organization also reduces the energy consumption. Fig. 20 shows the normalized energy

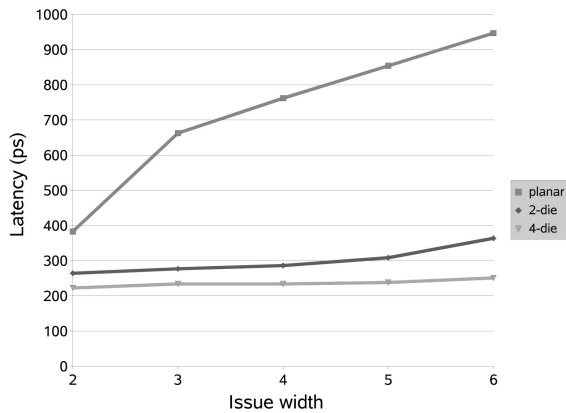


Fig. 19. Access latencies of a 96-entry, 64-bit register file for superscalar processors with increasing issue widths.

(normalized with respect to the corresponding planar array) required to access a data value from 2-die stacked register files. The 3D configuration that minimizes energy consumption is not necessarily the configuration that has the lowest latency. Note that for smaller number of entries, BP organization requires the least energy. As the number of entries increases above 64-entries, RP organization provides the most benefit. The RP approach effectively halves the bitline length for each doubling of the number of stacked die. The shorter bitlines greatly reduce the loading on the sense amplifiers, which, in turn, may be sized smaller to consume even less energy.

Fig. 21 shows the normalized energy required to access a data value from 4-die stacked register files. In the case of 4-die stacks, either RP-RP or BP-RP organizations provide the most energy benefits among all the configurations.

7 CONCLUSION

3D technology can provide significant benefits in terms of both performance and energy consumption. We demonstrated how 3D technology can be applied to the design of SRAM components to reduce the lengths of critical wires. We proposed designs of 3D-integrated SRAMs in case of both large, banked and small, multiported SRAM structures and their hybrid combinations. In each case, the reduction of the

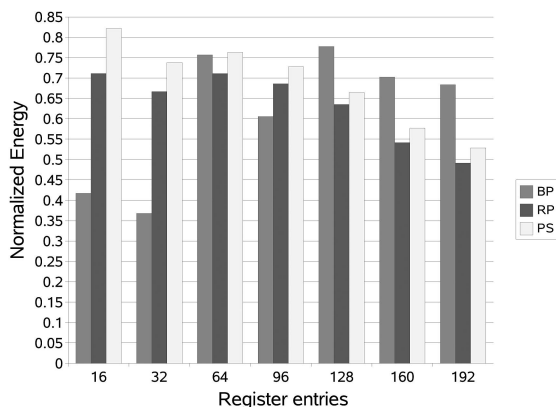


Fig. 20. Normalized energy (normalized to planar baseline) of 2-die stacked 3D register file designs.

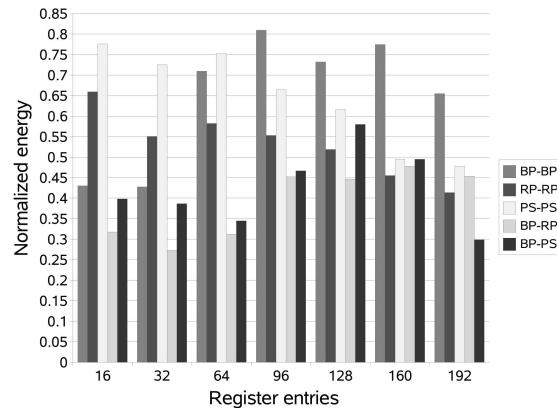


Fig. 21. Normalized energy (normalized to planar baseline) of 4-die stacked 3D register file designs.

wire length leads to substantial improvement in both latency and energy characteristics of the 3D SRAM components. As planar SRAM designs adapt high-performance techniques to improve performance, 3D integration provides even higher benefits, making it a desirable technology for high-performance designs. In addition to the SRAM components, many other components in modern processors are dominated by wire delay and benefit from a 3D implementation. Apart from adapting conventional microarchitectures to a 3D technology, designing a new processor from the ground-up to exploit 3D may provide even greater performance and power benefits. We believe that 3D integration holds tremendous promise and potential in addressing the challenges facing the microprocessor industry.

ACKNOWLEDGMENTS

Funding and equipment for this project have been provided by Intel Corporation and a grant from the Microelectronics Advanced Research Corporation (MARCO). The work was done when K. Puttaswamy was with the Department of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, 30332.

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